Behaving Sustainability: Cultural Cognition, Environmental Orientation, and Pro-Environmental Behavior

Introduction

Achieving environmentally sustainable outcomes requires some behavioral changes at the individual, community, and societal levels. However, not all individuals are equally likely to engage in pro-environmental behaviors, and understanding the psychological and cultural factors that drive these behaviors is important for enhancing sustainability. Scholars of environmental psychology often lament the so-called 'attitude-behavior' gap wherein environmental attitudes do not explain very much of the variance in environmental behavior (Gifford and Nilsson 2014; Kollmuss and Agyeman 2002; Stern 2000). If people care strongly about the environment, why don't they act in a more sustainable way? The attitude-behavior gap is not because people are hypocrites, but because researchers do not yet have a complete attitude-behavior model (Kaiser, Kibbe, and Hentschke 2021).

To understand how attitudes influence behaviors, scholars use approaches such as the Value-Belief-Norm (VBN) framework, where higher-order values shape second-order beliefs that, in turn, influence behaviors []. Higher-order values form a cognitive foundation that influences how information about the world is processed and understood across multiple issues []. When considering environmental behaviors, the second-order values involve environmental cognition, such as views about the relationship of humans to the natural world and how information about the environment is understood []. In addition to environmental cognition, there are also affective feelings that individuals hold about the environment []. Together environmental cognition and affective feelings form an environmental orientation that shapes environmental behaviors []. However, to develop a complete attitude-behavior model, it is important to understand all of the various links in the chain, including the higher-order values and cognition that inform environmental orientations. Of note, the cultural worldviews measured by cultural cognition have been shown to shape environmental orientations (Nowlin 2019).

Cultural cognition was developed by Kahan and others (Kahan et al. 2007) as a tool to measure how cultural values shape beliefs about risk. It is based on cultural theory and its conceptualization of worldviews []. Cultural theory addresses how cultural values influence how people perceive and respond to risk. By measuring the cultural values of an individual, we may be able to predict how they will perceive and respond to risks associated with certain activities or products. We are interested in how cultural cognition shapes views toward environmental issues and how those views shape behavior.

The cultural worldviews associated with cultural cognition are measured in two dimensions, a grid dimension (egalitarianism-hierarchy) and a group dimension (communitarianism-individualism). In general, those that are more egalitarian and communitarian are more concerned about environmental risks than those that are more hierarchical and individualist. Therefore, it is expected that individuals high in egalitarianism and communitarianism should engage in pro-environmental behaviors to mitigate risks to the natural environment. However, previous research shows only a weak correlation between egalitarianism and communitarianism and pro-environmental behavior (Sparks et al. 2021). Therefore, cultural cognition shapes how individuals think and process information about issues, but may not directly influence behaviors.

In this paper, we develop a theoretical framework that accounts for both the cognitive and affective influences on pro-environmental behavior. Specifically, we posit that that higher-order beliefs, drawn from cultural cognition (Kahan 2012; Kahan et al. 2012; Kahan and Braman 2006), inform environmental orientations (Henry and Dietz 2012), which in turn influence behavior, both public and private. Drawing from Henry and Dietz (2012), we use the New Ecological Paradigm (NEP) as our measure of environmental cognition, and the Connectedness to Nature Scale (CNS) as the affective element of environmental orientation. Previous work failed to find that cultural cognition influence environmental behavior; however, it has been found to influence environmental cognition using the NEP (Nowlin 2019). Therefore, we hypothesize that environmental orientations mediate the relationship between cultural cognition and behavior.

Overall we find ...

Next, we develop our approach based on previous literature.

Cognition, Environmental Orientations, and Pro-Environmental Behavior

The process though values and beliefs impact behavior is often understood as a process where higher-order beliefs influence second-order beliefs, which in turn influence behavior. One of the leading frameworks that adopts this approach is the Value-Belief-Norm (VBN) framework (Stern et al. 1999). In brief, the VBN posits that underlining values inform broad environmental orientations that activate norms that in turn support pro-environmental behavior. VBN involves a sequential chain that goes from values to beliefs to behavior. Therefore, values inform attitudes, which in turn inform behaviors.

Value-Belief-Norm (VBN) framework literature

A similar set of assumptions inform a sophisticated understanding of belief systems developed by policy scholar using the Advocacy Coalition Framework (ACF) (Henry and Dietz 2012; Nohrstedt et al. 2023; Sabatier and Jenkins-Smith 1993). With this approach, beliefs systems are made up of three different types of beliefs, including deep core, policy core, and secondary beliefs. Deep core beliefs are the broadest in scope and are fundamental beliefs that inform attitudes about multiple issues. Political ideology is an example of a deep core belief. Next, are policy core beliefs, which are beliefs about particular issues. Environmental orientations would be akin to policy core beliefs. Finally, secondary aspects are the narrowest in scope and include beliefs about how best to achieve the goals outlined in policy core beliefs.

In developing our approach, we draw on the notion of a sequential chain and a hierarchical ordering of values, beliefs, and attitudes that lead from abstract values to concrete actions. Specifically, we note that deep core beliefs inform environmental orientations that, in turn, make pro-environmental behavior more (or less) likely. In addition, we separate beliefs that shape *cognition* from the affective and identity elements of environmental orientations. Our measure of deep core beliefs is cultural cognition, which as the name implies, are deep rooted values that influence cognition. For environmental orientations, we draw on environmental cognition, which is "the way individuals structure their thinking about environmental issues and associated political actions" (Henry and Dietz 2012, 238). We argue that environmental orientation contain a cognitive element as well as affective element. Deep core beliefs influence both the cognitive and affective aspects of environmental orientation. Finally, deep core beliefs and environmental orientations influence pro-environmental behaviors; however, deep core beliefs likely influence behaviors *indirectly* [], with the affective element of environmental orientations having the most direct influence on behaviors [].

that higher-order beliefs, like deep core beliefs, shape how people think about issues and filter information

cultural and environmental cognition.

Work using environmental cognition

belief system framework of the ACF

cultural and environmental cognition

Policy scholars have developed a that can inform how beliefs influence attitudes []. While not much work has connected belief systems to behavior

belief systems

Environmental orientations are part of broader belief systems, and policy scholars using the Advocacy Coalition Framework have developed a three-tiered hierarchical model of belief systems

Cultural theory, developed by Mary Douglas and Aaron Wildavsky

Deep core beliefs

Additional work has examined the importance of belief systems, as developed by [].

Environmental cognition connects the belief system of the ACF to VBN

In this paper, we build on this approach by developing a theoretical framework

The beliefs associated with VBN are often measured using the New Ecological Paradigm (NEP). However, recent work found the New Ecological Paradigm (NEP) scale plus controls accounted for 12% of variation in self-reported pro-environmental behavior (Sparks et al. 2021).

and the Connectedness to Nature Scale (CNS) plus controls accounted for 35%.

To further elaborate the attitude-behavior model we examine the relationship between cultural cognition, environmental orientation (attitude and emotional connection), and behavior.

; however, cultural cognition does not have a strong association with pro-environmental behaviors (Sparks et al. 2021).

In this paper, we examine whether environmental orientations mediate the relationship between cultural cognition and behavior such that cultural cognition shapes environmental orientations, which in turn influences pro-environmental behavior. In doing so, we argue that when cultural cognition aligns with environmental orientation, there is a stronger relationship with pro-environmental behaviors than when considering attitude or affect alone, thus reducing the attitude-behavior gap.

Taking the step from cultural worldviews to environmental orientation , we test the relationship between cultural cognition and environmental attitude – as measured by the New Ecological Paradigm [NEP; Dunlap et al. (2000)] – and emotional connection to nature – as measured by the Connectedness to Nature Scale [CNS; Mayer and Frantz (2004)]. Altogether, combining cultural cognition and environmental orientation should improve our understanding of the drivers of environmentally protective behaviors. Specifically, in this paper, we suggest that environmental orientation mediates the relationship between cultural cognition and environmental behavior. Essentially, environmental orientation helps connect cultural worldviews and risk perceptions to action to reduce that risk. Moreover, cultural cognition helps to bridge the attitude-behavior gap by providing a basis through environmental orientations derive and subsequently influence behavior.

Cultural Cognition

Cultural cognition is derived from the cultural theory of risk developed by Mary Douglas and Aaron Wildavsky (Douglas and Wildavsky 1982; Thompson, Ellis, and Wildavsky 1990).

In brief, cultural cognition posits that worldviews are based on views about ideal social structures and relationships, and perceptions of risk are based on the degree to which they undermine or threaten the preferred social order within each worldview.

The social structures and relationships are represented by two dimensions including a grid dimension and a group dimension. Four competing cultural worldviews arise based on their position on the grid and group dimensions. The grid dimension represents preferred social structures regarding the roles that one plays in society and the rules by which one must abide. The worldview high on the grid dimension prefers a hierarchy in society with well-defined roles and clear lines of authority. Egalitarianism is the worldview low on the grid dimension and values a social order based on an equal sharing of power and resources. The group dimension represents the importance of cohesive social units and the well-being of the broader community. Communitarianism is the worldview that is high on the group dimension, and it represents a preference for a collectivist society. Finally, individualism is low on the group dimension preferring increased individual autonomy and responsibility (Kahan 2012; Kahan and Braman 2006).

Cultural worldviews provide a foundational, or core belief that inform beliefs about specific policy issues, societal controversies, and risks, among other beliefs (Nowlin 2019; Ripberger et al. 2014). Several studies have shown a connection between cultural worldviews and risk perceptions associated with a host of environmental issues (see Xue et al. 2014) including fracking (Boudet et al. 2014), nuclear waste (Jenkins-Smith and Smith 1994), renewable energy (Tumlison et al. 2018), geoengineering (Kahan et al. 2015), high-voltage power lines (Moyer and Song 2016, 2019), and climate change (Jones 2011; Kahan et al. 2012; Nowlin and Rabovsky 2020; Price, Walker, and Boschetti 2014), among others.

In general, communitarianism, with its concern about collective well-being and egalitarianism, with its concern about the negative impacts of concentrated power tend to see threats to the natural world as a large risk. Hierarchs believe that natural resources can be well managed by experts and individualism supports the use of natural resources in private enterprise, and therefore hierarchy and individualism tend not to be concerned about environment harm. As noted, cultural worldviews have been found to be associated with attitudes about environmental issues, yet not associated with pro-environmental behaviors (Sparks et al. 2021). However, cultural cognition is associated with pro-environmental orientation, specifically the New Ecological Paradigm (NEP) (Nowlin 2019).

New Ecological Paradigm

To understand environmental behavior, scholars often turn to mental attributes related to environmental orientation. A very common measure of environmental orientation is the New Ecological Paradigm (NEP) (Dunlap et al. 2000). This instrument was initially developed by Dunlap and Van Liere in 1978 and has been widely adopted to explain environmental behavior (e.g., Stern 2000). In 2000, Dunlap et al revised the NEP.

The 1978 conceptualization focused on what they considered an emergent environmental paradigm that challenged the dominant social paradigm (Dunlap and Van Liere 1978). Dunlap and co-authors (2000) re-imagined the NEP in 2000, shifting focus to an ecological worldview and toward more modern language to describe the ecological crises. The new measure comprises five aspects of this ecological paradigm which are "the reality of the limits to growth, anti-anthropocentrism, the fragility of earth's balance, rejection of exemptionalism, and the possibility of an ecocrisis" (p. 432). To assess this construct, they designed 15 Likert items that contain a mix of attitude and belief measures which are balanced between a pro-environmental orientation and a negative environmental orientation (Dunlap et al. 2000).

The NEP is not without skeptics. One study applying scale analysis of the NEP based on a random sample of Norwegians found that the scale is not unidimensional as is often claimed (Grendstad 1999). A recent study found the NEP to be the weakest predictor of environmental behavior when compared in the same sample to the Connectedness to Nature scale and environmentalist identity (Sparks et al. 2021). This may be because endorsement of the NEP has decreased over time (Lou et al. 2022). In another study, Sparks and his colleagues (2022) criticize the NEP (and the CNS) for failing to adequately identify political conservatives who also hold a pro-environment orientation.

Connectedness to Nature

A second common instrument measuring environmental orientation is the Connectedness to Nature Scale (CNS) (Mayer and Frantz 2004). As implied by its name, the CNS seeks to measure an individual's emotional connection to nature, rather than environmental attitude. Mayer and Franz draw inspiration from Leopold's insistence that to address environmental problems, people must feel emotionally connected to the non-human community on which we are dependent upon. The idea is that current ecological crises are a direct result of humanity losing touch with the broader community in which we live. The hypothesis can be tested by measuring an individual's affective connection to nature and determining if someone with more emotional connection correlates to more environmentally protective behaviors. They demonstrate this in their study and show that the CNS better predicts behavior (Mayer and Frantz 2004). The full instrument is composed of 14 Likert items designed to directly assess an individual's affective connection to nature.

Going from scale development, to specific applications, one study assessed the role of personality traits on environmental behavior (Brick and Lewis 2016). The study also provided a comparison of the mediation effect of both the NEP and the CNS on environmental behaviors. The personality traits Openness and Conscientiousness were both shown to be positive predictors of environmental behavior. In both cases, the CNS had a stronger mediation effect than NEP. In other words, the CNS better predicts pro-environmental outcomes. However, these authors also note that the CNS contains language that is more consistent with liberal environmentalism, and so may fail to adequately capture the full dimensionality of environmental support across the political spectrum (Brick and Lewis 2016).

Like the NEP, the CNS is composed mostly of items that tap into politically progressive environmentalism and thus may miss conservative pro-environmental orientation (Sparks, Ehret, and Brick 2022). However, the CNS is a much stronger predictor of environmental behavior – a smaller attitude-behavior gap – than the NEP (Sparks et al. 2021).

Pro-Environmental Behaviors

Cultural cognition can be thought of as a determinant or antecedent of environmental orientation. Cultural views of risk and social-psychological tendencies likely precede the development of environment-specific attitudes and orientations. And given the weak correlation between cultural cognition and environmental behavior it is reasonable to view this as an indirect relationship.

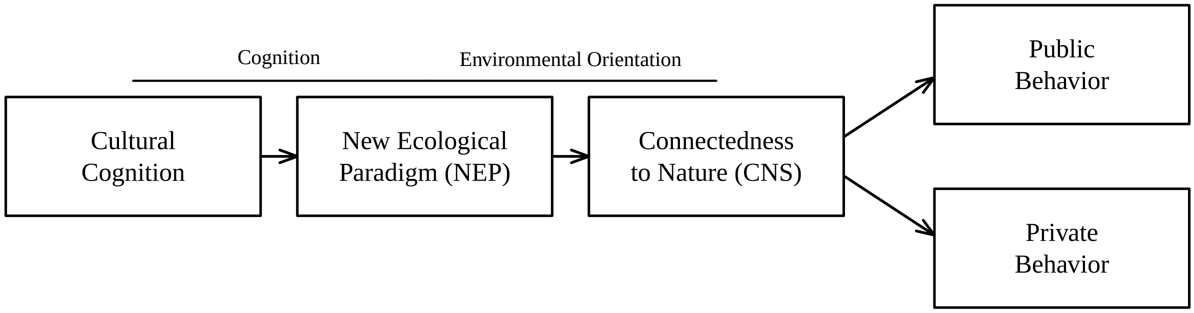
Worldview and risk perception are likely strong drivers of environmental orientation because of the nature of environmental problems typically involves environmental risk. In addition, environmental orientation is likely associated with the myths of nature connected to the various cultural worldviews. As a result, environmental orientation, as measured by NEP and/or CNS, likely mediates the relationship between cultural cognition and environmental behavior. Figure 1 illustrates the conceptual model connecting cultural cognition, environmental orientation, and pro-environmental behavior.

Hypotheses

Based on the model shown in Figure 1, we posit several hypotheses regarding the relationships between cultural cognition, the NEP, the CNS, and environmental behavior.

As noted, egalitarianism and communitarianism has been shown to be predictive of concern regarding several environmental issues and environmental orientations. Therefore, we expect that:

Figure 1: Conceptual model tested with the structural equation model: cultural cognition shapes environmental orientation—first the New Ecological Paradigm (NEP) and then the Connectedness to Nature Scale (CNS)—which in turn shapes public and private pro-environmental behavior.



H1a: Egalitarianism positively correlates with environmental behavior H1b: Egalitarianism positively correlates with environmental orientation (NEP and CNS)

H2a: Communitarianism positively correlates with environmental behavior H2b: Communitarianism positively correlates with environmental orientation (NEP and CNS)

Previous research indicates that cultural cognition predicates environmental orientation and environmental orientation predicts pro-environmental behavior. Therefore, we posit that:

H3: Environmental orientation positively mediates the relationship between egalitarianism and environmental behavior

H4: Environmental orientation positively mediates the relationship between communitarianism and environmental behavior

Additionally, through meditation analysis, we can examine the direct, indirect, and total effects of cultural cognition and the mediating effects of environmental orientation on environmental behavior.

Data and Methods

To examine the relationship between cultural cognition, environmental orientations, and behavior we draw on a sample of 501 adults living in the US purchased through Survey Sampling Inc. The survey was administered online in July 2017. The sample size was determined by cost considerations. IRB approval for the survey was approved by [redacted for blind review] Human Subjects Committee on April 3, 2017, protocol number 37-17-0256. Our analysis was not pre-registered.

Measures

Our dependent variable is a 13-item scale of self-reported pro-environmental behaviors with the questions taken from the 2000 Gallup Earth Day Poll. Respondents were given a list of several environmental behaviors and asked, "Which of these, if any, have you, yourself, done in the past year?" Possible responses included, "Yes, have done; No, have not done; No opinion." Yes, responses were coded as a 1 and no/no opinion were coded as 0. For the purposes of analysis, we treat the behavior measure as observed, and we add the number of self-reported behaviors together to produce a measure that ranges from 0 (no behaviors) to 13 (all of the behaviors). The mean number of self-reported behaviors was 5.52, with a standard deviation of 2.98. Table 1 shows the wording for each behavior as well as the mean and sd for each self-reported behavior.

Table 1: Self-reported pro-environmental behaviors by type (public/private), with means and standard deviations

Behavior	Type	Mean	SD
Been active in a group/organization that works to protect the environment	Public	0.19	0.39
Voted for or worked for candidates because of their environmental positions	Public	0.40	0.49
Contributed money to an environmental, conservation, or wildlife group	Public	0.33	0.47
Contacted a public official about an environmental issue	Public	0.19	0.39
Contacted a business to complain about environmentally harmful products/policies	Public	0.16	0.36
Signed a petition supporting environmental protection	Public	0.36	0.48
Attended a meeting concerning the environment	Public	0.17	0.37
Public behaviors (0-7)		1.80	1.92
Avoided using products that harm the environment	Private	0.66	0.47
Tried to use less water in your household	Private	0.75	0.43
Bought a product because it was better for the environment	Private	0.62	0.49
Voluntarily recycled newspapers, glass, aluminum, motor oil, etc.	Private	0.79	0.41
Reduced your household's use of energy	Private	0.78	0.42
Private behaviors (0-5)		3.60	1.58
Total behaviors (0-12)		5.40	2.86

As shown in Table 1, recycling had the highest mean among the self-reported behaviors at 0.79, which indicates that 79% of the respondents reported that they recycle. Recycling was followed closely by reducing household energy use at 0.78 and using less water at home at 0.75. The behaviors with the lowest means included buying/selling stocks at 0.12, contacting a business at 0.16, and attending a meeting at 0.17.

Cultural Cognition

To measure cultural cognition, we rely on the short form of the cultural cognition survey questions which includes three questions for each of the four cultural worldviews (see Kahan 2012). Respondents were asked their agreement on a 1 (strongly disagree) to 4 (strongly agree) scale with each of the following statements. The hierarchy statements included, 1) We have gone too far in pushing equal rights in this country, 2) It seems like blacks, women, homosexuals and other groups don't want equal rights, they want special rights just for them, and 3) The women's rights movement has gone too far. The egalitarianism statements are, 1) We need to dramatically reduce inequalities between the rich and poor, whites and people of color, and men and women, 2) Our society would be better off if the distribution of wealth was more equal, and 3) We live in a sexist society that is fundamentally set up to discriminate against women. The individualism statements included, 1) If the government spent less time trying to fix everyone's problems, we'd all be a lot better off, 2) Government regulations are almost always a waste of everyone's time and money, and 3) Our government tries to do too many things for too many people. We should just let people take care of themselves. Finally, the communitarianism statements are, 1) Sometimes government needs to make laws that keep people from hurting themselves, 2) Government should put limits on the choices individuals can make so they don't get in the way of what's good for society, and 3) The government should do more to advance society's goals, even if that means limiting the freedom and choices of individuals.

New Ecological Paradigm

To measure the New Ecological Paradigm (NEP) we used agreement with the following three statements on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree). The statements are, 1) Humans are severely abusing the environment, 2) The so-called "ecological crisis" facing humankind has been greatly exaggerated (reverse-coded), and 3) If things continue on their present course, we will soon experience a major ecological catastrophe.

Connectedness to Nature

We measured the Connectedness to Nature Scale (CNS) using respondent agreement on a 1 to 5 scale (1 - strongly disagree, 3 - neutral, 5 - strongly agree) with three statements. The statements included, 1) I think of the natural world as a community to which I belong, 2) I often feel a kinship with animals and plants, and 3) I feel as though I belong to the Earth as equally as it belongs to me.

Table 2 shows the measurement variables used for the latent constructs including the mean and standard deviation (sd) of each statement. In addition, Table 2 show the fit statistics for each of the latent constructs including Cronbach's α, Composite Reliability (CR), and the Average Variance Extracted (AVE).

Table 2: Measurement items and construct reliability (Cronbach's α, composite reliability, average variance extracted)

Construct	Item	Mean	SD	α	CR	AVE
Hierarchy	cc_eh_2	2.33	1.01	0.83	0.83	0.62
	cc_eh_5	2.43	1.08			
	cc_eh_7	2.16	0.98			
Egalitarianism	cc_eh_11	3.13	0.88	0.77	0.79	0.55

Table 2: Measurement items and construct reliability (Cronbach's α, composite reliability, average variance extracted)

Construct	Item	Mean	SD	α	CR	AVE
	cc_eh_13	2.94	0.91			
	cc_eh_14	2.60	0.87			
Individualism	cc_ci_2	2.69	0.91	0.80	0.80	0.57
	cc_ci_3	2.53	0.90			
	cc_ci_12	2.58	0.86			
Communitarianism	cc_ci_13	3.01	0.70	0.61	0.64	0.38
	cc_ci_14	2.44	0.83			
	cc_ci_16	2.31	0.85			
CNS	cns2	3.78	0.90	0.76	0.72	0.47
	cns6	3.69	1.05			
	cns7	3.71	1.00			
NEP	nep5	3.99	1.14	0.70	0.73	0.48
	nep10	3.25	1.38			
	nep15	3.80	1.15			

As shown in Table 2, each of the constructs show sufficient reliability (α and CR >0.70) except for the communitarianism scale. However, CR values between 0.60 and 0.70 are acceptable for exploratory analysis (Hair et al. 2015).

To test our hypothesis, we estimated a Structural Equation Modeling (SEM) using the `lavaan` package in `R` (Rosseel 2012). The dependent variable, self-reported pro-environmental behaviors, is an observed variable and the cultural cognition worldviews, the CNS, and the NEP scale are latent constructs. We run separate analyses for the two measures of environmental orientation, CNS and NEP, and for both models we show the standardized regression coefficients for the measurement and structural models. In addition, we assess the fit of both models using the Root Mean Square Error of Approximation (RMSEA), the Comparative Fit Index (CFI), and the Standardized Root Mean Square Residual (SRMR). Finally, we examine the mediation effects of environmental orientation by examining the direct, indirect, and total effects of cultural cognition measures on pro-environmental behavior.

Results

Table 3: Fit comparison of the strict serial chain versus the full combined model with direct paths added

Model	Chi-sq	df	CFI	TLI	RMSEA	RMSEA 90% CI	SRMR	AIC	BIC
Serial chain	499.29	158	0.902	0.882	0.066	[0.059, 0.072]	0.071	25570.1	25789.4

Table 3: Fit comparison of the strict serial chain versus the full combined model with direct paths added

Model	Chi-sq	df	CFI	TLI	RM-SEA	RMSEA 90% CI	SRMR	AIC	BIC
Full (+ direct paths)	420.63	144	0.921	0.895	0.062	[0.055, 0.069]	0.061	25519.4	25797.7

Nested χ^2 difference test (serial chain nested in full model): $\Delta\chi^2(14) = 78.66, p = 0.0000$. The direct paths jointly improve fit.

Table 4: Standardized decomposition of cultural cognition effects on public and private behavior in the full combined model († p<0.10, * p<0.05, ** p<0.01, *** p<0.001)

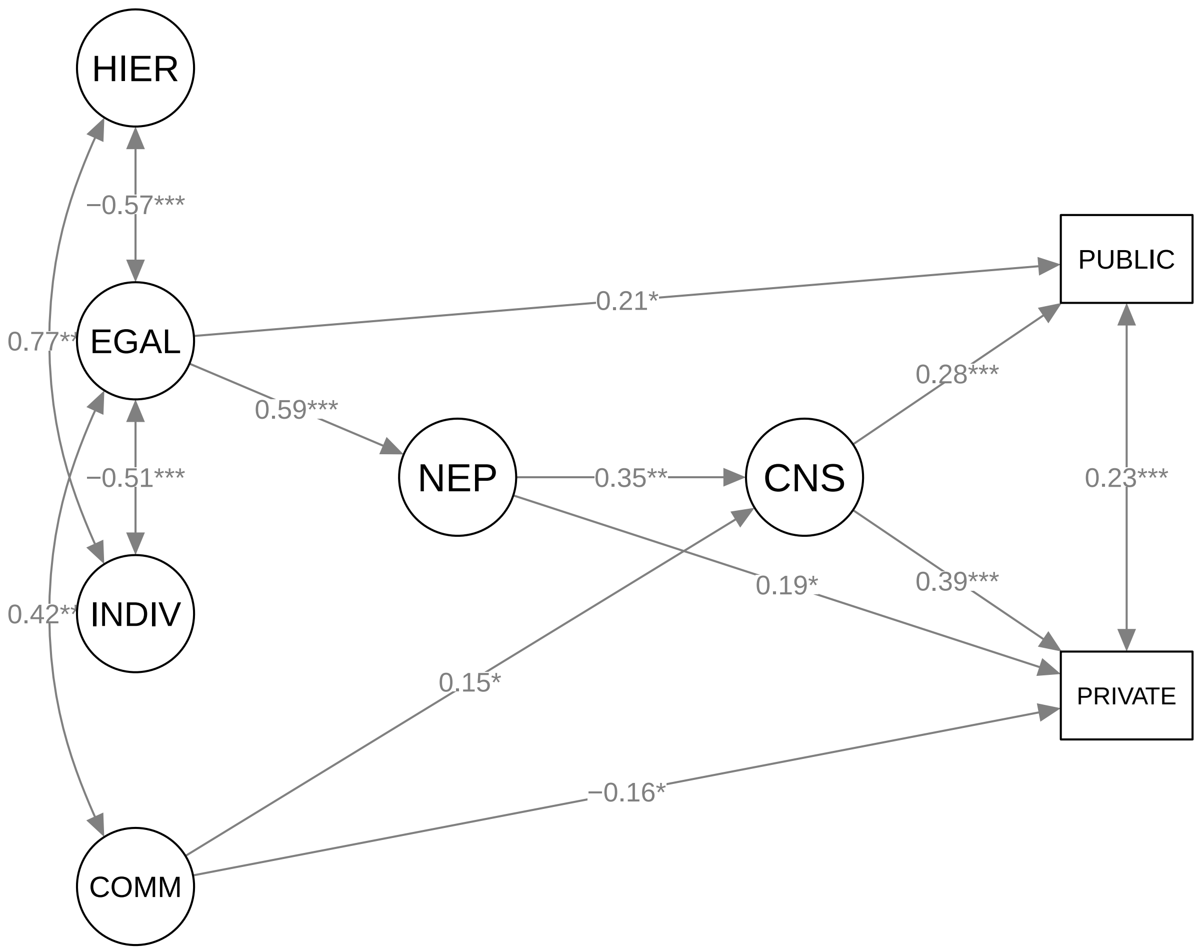
Predictor	Outcome	Direct	Serial (NEP→CNS)	Total indirect	Total
Egalitarianism	Public	0.205*	0.057*	0.155**	0.360***
Egalitarianism	Private	-0.084	0.079**	0.222***	0.138†
Communitarianism	Public	0.123†	0.006	0.057*	0.181**
Communitarianism	Private	-0.163*	0.009	0.081*	-0.082
Hierarchy	Public	0.121	-0.014	-0.075†	0.046
Hierarchy	Private	-0.130	-0.020	-0.106*	-0.236*
Individualism	Public	0.080	-0.009	0.026	0.107
Individualism	Private	0.016	-0.013	0.036	0.052

Discussion

Scholars have noted a gap between attitudes and beliefs about the environment and pro-environmental behavior. Additionally, cultural worldviews derived from cultural cognition, specifically egalitarianism and communitarianism, have been shown to inform views about several environment issues as well as overall attitude toward the environment. But previous research has not found a connection between cultural worldviews and environmental behavior. In this paper, we developed and tested a conceptual model, based on previous research, where cultural worldviews influence environmental orientations and behaviors, and environmental orientations influence behaviors. Additionally, we examined the mediation effects of environmental orientations on the relationship between cultural cognition and pro-environmental behaviors. Environmental orientations were measured in terms of attitudes, using the NEP, and the emotional connection to nature, using the CNS. We used SEM to test each of our hypotheses.

We posited that egalitarianism and communitarianism would be positively associated with environmental orientations and environmental behavior (H1 a and b, H2 a and b). We found mixed results for our hypotheses. As expected, egalitarianism is associated with both the CNS and NEP; however, egalitarianism was only associated with behavior in the CNS model. However, communitarianism was associated with environmental orientation only in the CNS model and not associated with behaviors in either model.

Figure 2: Structural model of the full combined SEM: cultural cognition → New Ecological Paradigm → Connectedness to Nature → public and private pro-environmental behavior, with direct paths. The measurement (indicator) portion is omitted for clarity; standardized coefficients shown; * p<0.05, ** p<0.01, *** p<0.001. The full path diagram including the measurement model is in the supplemental materials.



Additionally, we posited that environmental orientations mediate the relationships between egalitarianism (H3) and communitarianism (H4) and pro-environmental behaviors. As expected, both the NEP and the CNS significantly mediate the relationship between both egalitarianism and communitarianism, and environmentally protective behavior. As far as we know, this is a novel finding. While previous studies have shown a relationship between cultural cognition and environmental orientation (Nowlin 2019) none have included a mediation analysis. Along with the results of the structural equation modeling, this helps us understand cultural cognition as an antecedent of environmental orientation and fill some of the attitude-behavior gap.

We also found that the Connectedness to Nature Scale likely had more predictive power regarding environmental behaviors than the NEP based on the model fit statistics in Table 3. This finding is in line with work showing that the CNS is a stronger predictor than the NEP (Sparks et al. 2021).

There are at least two potential reasons for the CNS displaying more power than the NEP to influence behavior, which we are only able to speculate about in the present study. First, prior work has demonstrated a slow decline in environmental concern as measured by the NEP (Lou et al. 2022). Lou and colleagues do not provide evidence for why this may be happening, only show through meta-analysis this downward trend. We are not aware of a similar study tracking CNS over time, but that may prove a fruitful avenue for further study. If it is only the NEP that is declining over time, that may indicate that the measurement instrument is becoming outdated, potentially because of shifts over time making the distinction between the dominant social paradigm and the new ecological paradigm less apt. For example, many see technological innovation as the key to unlocking a new green economy, rather than destroying the earth. Relatedly, perhaps the language used in the NEP has just become outmoded and out of line with how environmentalists currently would express environmental attitude (see Sparks, Ehret, and Brick 2022 for a discussion of this).

Second, this may be a result of the CNS and NEP measuring two distinct concepts that can be grouped under the umbrella of environmental orientation. The NEP is explicitly a measure of environmental-attitude while CNS is a measure of emotional connection to nature. For example, a recent study found that harnessing the right mix of emotions is a powerful motivator of climate action (Kovács et al. 2024).

Conclusion

In this paper, we examined the relationship between cultural cognition, environmental orientation, and pro-environmental behavior. We presented evidence that cultural cognition, egalitarianism not communitarianism, is best thought of as an indirect predictor of pro-environmental behavior. The relationship was mediated by pro-environmental orientation as measured by both the NEP and the CNS.

Because cultural cognition shapes how an individual responds to risk (e.g., Kahan et al. 2012), it was something of a puzzle as to why it only weakly correlates to behavior that protects the environment – an action that can be seen as a way to reduce environmental risk to the individual. The evidence we presented in this paper suggests that environmental orientation – whether attitude (NEP) or emotional connection (CNS) – is a missing link in this relationship.

Further, by explaining behavior with a mediation model we could close some of the so-called attitude-behavior gap. When cultural cognition and environmental orientation are aligned, people are more likely to engage in environmentally protective behavior.

Another implication of this study is further evidence that CNS is a stronger predictor of behavior than NEP. This may be useful for researchers trying to decide which measure of pro-environmental orientation to include in their questionnaire. We would like to see further testing of this finding in studies in other contexts and through meta-analytic techniques.

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